

Current Electricity

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Q1. A current of 10 A exists in a wire of cross-sectional area of 5 mm² with a drift velocity of 2×10^{-3} m/s. The number of free electrons in each cubic metre of the wire is ____.

JEE Main (Physics, Current Electricity)

✓ **Answer: 6.25×10^{27} per m³**

Solution:

Using $I = neAv_d$, solve for n:

$$n = I/(eAv_d) = 10/[(1.6 \times 10^{-19})(5 \times 10^{-6})(2 \times 10^{-3})]$$
$$= 10/(1.6 \times 10^{-27}) = 6.25 \times 10^{27} \text{ per m}^3.$$

Q2. A resistor develops 500 J of thermal energy in 20 s when a current of 1.5 A is passed through it. If the current is increased from 1.5 A to 3 A, what will be the energy developed in 20 s?

JEE Main (Physics, Current Electricity)

✓ **Answer: 2000 J**

Solution:

Heat produced $H = I^2Rt$, so for the same R and t, $H \propto I^2$.

$$\text{Ratio of energies} = (I_2/I_1)^2 = (3/1.5)^2 = 4.$$

$$\text{New energy} = 500 \times 4 = 2000 \text{ J}.$$

Source: JEE Main 2026 (NTA official releases, physics section) · formulaverse.in